

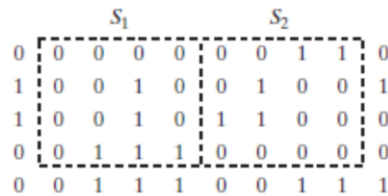
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) What are the components of Digital Image Processing? Explain in detail. (BT2/CO1) **6**
- B) Consider the two image subsets, S1 and S2, shown in the following figure. For $V=\{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent. (BT3/CO1) **6**



- C) Differentiate Image Processing & Computer Vision. Explain the application of Image Processing and Computer Vision. (BT2/CO1) **6**

Q.2 Solve Any Two of the following.

- A) What is the image histogram? Explain histogram equalization and matching. (BT2/CO2) **6**
- B) Compute Convolution and correlation of $f(x, y)$ and $w(x, y)$. (BT4/CO2) **6**

$$f(x, y) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } w(x, y) = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- C) Compute DFT of $f(x, y)$ (BT3/CO2) **6**

$$f(x, y) = \begin{bmatrix} 2 & 4 & 4 & 2 \\ 4 & 6 & 8 & 3 \\ 2 & 8 & 10 & 4 \\ 3 & 8 & 6 & 2 \end{bmatrix}$$

Q. 3 Solve Any Two of the following.

- A) Define and explain Hit & Miss Transform with suitable example. (BT1/CO3) **6**
- B) Discuss the Hole filling algorithms in brief with examples. (BT2/CO3) **6**

C) Compute dialation and erosion of A & B.

(BT4/CO1)

6

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Q.4 Solve Any Two of the following.

A) Explain Polygonal Approximation with suitable example.

(BT1/CO4)

6

B) What do you mean by Feature descriptor? Explain shape number.

(BT2/CO4)

6

C) Explain Point and Line Detection in segmentation.

(BT2/CO4)

6

Q. 5 Solve Any Two of the following.

A) Illustrate Pattern and pattern classes.

(BT2/CO5)

6

B) Write note on Neural Network and Deep Learning.

(BT3/CO5)

6

C) Explain Bayes statistical classifiers.

(BT4/CO5)

6

***** End *****